

Project Documentation: Sentiment & Progress Analytics

End-to-End Data Engineering & Business Intelligence Workflow

1. Executive Summary

This project showcases a sophisticated data lifecycle management process, transitioning from raw unstructured data to high-impact business visualizations. By analyzing stakeholder sentiment across the **Banking**, **MEA**, and **MoRTH** sectors, the solution provides a data-driven framework for operational improvement and public relations management.

2. Technical Architecture & Workflow

Step 1: Data Synthesis & Simulation

To demonstrate high-level technical skills, data was initially simulated using **Google Gemini**. This data was structured into an HTML format to mimic real-world web environments, setting the stage for advanced data extraction techniques.

Step 2: Web Scraping (Python / Colab)

Utilizing **Python** in a Google Colab environment, a specialized scraping script was executed. This phase involved:

- Parsing complex HTML structures.
- Extracting tabular data into a structured data frame.
- Exporting results into a .csv format for downstream processing.

Step 3: ETL & Data Refinement (Excel)

The raw CSV underwent a rigorous **Extract, Transform, Load (ETL)** process in Excel to ensure data integrity:

- **Type Correction:** Standardized date formats for time-series compatibility.
- **Cleaning:** Systematic removal of duplicates and handling of null values.
- **Normalization:** Transformation of percentage-based "Impact Weight" strings into decimal/numeric values to allow for weighted average calculations.

3. Data Structure Overview

Column Name	Technical Role	Business Value
Record_ID	Primary Key	Ensures data traceability.
Date/Month	Time Dimension	Enables trend and seasonality analysis.
Sector Entity	Category	Segments data by department (Banking, MEA, MoRTH).
Sentiment	Measure	Quantifies emotional tone (Positive, Neutral, Negative).
Impact Weight	Weighting Factor	Prioritizes high-stakes feedback.

4. Power BI Visual Analysis

The dashboard is engineered to solve specific organizational challenges through targeted visualizations:

Sentiment Distribution: Identifies the immediate public health of the brand. High negative ratios trigger immediate PR intervention.

Sector-Wise Analysis: Pinpoints departmental friction. If the "Banking Sector" shows lower sentiment than "MoRTH," it signals a need for service delivery audits in finance.

Aspect Category Heatmap: Drills down into the why. Identifies if negativity is tied to "Visa Services" (operational) vs "Interest Rates" (macro-economic).

Temporal Trends: Correlates sentiment shifts with external events or policy launches, providing a feedback loop for leadership.

5. Conclusion

This pipeline demonstrates proficiency in **Python automation**, **Data Engineering (ETL)**, and **Strategic Data Visualization**. The final Power BI dashboard serves as a "Single Source of Truth," enabling stakeholders to move from reactive management to proactive, data-informed strategy.